

team collaboration and streamline the development workflow.

Continuous integration: Git integrates seamlessly with CI tools like Jenkins, CircleCI, and Travis CI. Automated build and test processes can be triggered by code changes, ensuring that new code is tested and integrated continuously.

Building DevOps pipelines with open source tools

Open source tools play a pivotal role in building and managing DevOps pipelines. These tools help automate testing, deployment, and monitoring, ensuring that software is delivered rapidly and reliably.

Jenkins: Jenkins is an open source automation server that facilitates CI/CD. Its extensive plugin ecosystem allows integration with various tools and services, making it highly adaptable to different workflows. Companies like Capital One use Jenkins to automate their CI/CD pipelines, ensuring that code changes are tested and deployed rapidly. Jenkins' plugin ecosystem allows integration with various tools and services, enhancing its functionality.

Ansible: Ansible is an open source automation tool for configuration management, application deployment, and task automation. Its simple, agentless architecture and human-readable YAML configuration files make it easy to use and integrate into DevOps pipelines.

Docker: Docker is a platform for developing, shipping, and running applications in containers. Containers provide a consistent environment across different stages of the development lifecycle, from development to production, ensuring that applications run reliably in any environment.

DevSecOps: Integrating security into DevOps

Incorporating security into every stage of the DevOps pipeline is critical.

DevSecOps emphasises the importance of continuous security monitoring, vulnerability management, and compliance, ensuring that security is an integral part of the development process.

The security pipeline involves integrating security practices into the CI/CD pipeline, ensuring that security checks are performed continuously and automatically.

Automated security testing: Tools like OWASP ZAP, Snyk, and Clair automate security testing, identifying vulnerabilities in code, dependencies, and container images. Automated testing helps detect and fix security issues early in the development process.

- **OWASP ZAP:** An open source security testing tool that automates vulnerability scanning for web applications. OWASP ZAP integrates with CI/CD pipelines, ensuring that security testing is performed continuously throughout the development lifecycle.
- **Snyk:** Focuses on identifying and fixing vulnerabilities in open source dependencies. By integrating with Git repositories and CI/CD tools, Snyk provides real-time security feedback and automated remediation.

Continuous monitoring: Security monitoring tools like Nagios and Snort continuously monitor applications and infrastructure for security threats. They provide real-time alerts and detailed reports, enabling quick response to potential security incidents.

- **Nagios:** Monitors infrastructure and applications for performance and security issues. Nagios' customisable alerts and dashboards provide real-time visibility into system health, enabling quick response to potential threats.
- **Snort:** This is an open source intrusion detection system (IDS) that analyses network traffic for suspicious activity. Snort's rule-based detection helps identify and mitigate security threats in real time.

Compliance and auditing:

DevSecOps practices include automated compliance checks and auditing to ensure that applications meet regulatory and security standards. Tools like OpenSCAP and AuditD help in maintaining compliance and generating audit reports.

Skills needed, certifications and career opportunities in DevOps

The demand for skilled DevOps professionals is on the rise, and acquiring the right skills and certifications can open up numerous career opportunities. DevOps professionals need a blend of technical and soft skills.

Scripting and automation:

Proficiency in scripting languages like Python, Bash, and PowerShell is essential for automating tasks and managing infrastructure.

Cloud platforms: Knowledge of cloud platforms like AWS, Azure, and Google Cloud is crucial for deploying and managing applications in the cloud.

CI/CD pipelines: Understanding CI/CD processes and tools is vital for automating software delivery and ensuring continuous integration and deployment.

Collaboration and communication:

Strong communication skills are necessary for collaborating with cross-functional teams and stakeholders.

Certifications validate expertise and enhance career prospects. Some of the most recognised DevOps certifications include:

- AWS certified DevOps engineer
- Certified Kubernetes administrator (CKA)
- Google professional DevOps engineer
- Docker certified associate (DCA)

Continuous learning through online courses, workshops, and hands-on experience is crucial for staying updated with the latest trends and technologies.

The demand for DevOps professionals is growing, with roles such as: